

Classical Still Wins the Wall-Clock: An Honest Quantum-vs-Classical Benchmark of Twenty Canonical Problems on Five Real Frameworks

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Abstract—Quantum computing is the most over-marketed corner of computing: tutorials stop at a toy circuit or imply an “advantage” that does not yet exist. This report does the opposite. It formulates twenty canonical quantum problems, attacks each with the real, dedicated frameworks (Qiskit with Aer, PennyLane, Cirq, Stim, and NumPy/scikit-learn for the classical baselines), and puts every quantum method next to its classical baseline with both costs on the table. The result, from 119 committed, reproducible traces, is unambiguous and honest: at lab scale the classical baseline is faster in wall-clock time on *all twenty* problems, usually by two to four orders of magnitude, and this holds even for the flagship “advantage” algorithms, Grover’s search and Shor’s factoring, because their asymptotic advantages do not materialise at the small instance sizes a laptop simulator reaches. Yet the genuine quantum phenomena are real and we show them: the CHSH value reaches the Tsirelson bound $2\sqrt{2}$ and violates the classical local-hidden-variable bound of 2 (but only with entanglement; a separable state gives 1.41), Grover uses provably fewer oracle queries (2 versus 7 on one instance), and Shor’s order-finding factors 15 correctly. The honest synthesis is that quantum genuinely wins at *physics* (nonlocality, query complexity) and does not yet win at *practical computation*: a quantum computer does not beat a classical one at anything you would pay for today, and this benchmark shows exactly why, with the numbers, over real frameworks, reproducibly.

Index Terms—Quantum computing, quantum advantage, benchmarking, Grover, Shor, CHSH, Bell inequality, QAOA, VQE, NISQ, reproducibility, honest evaluation.

I. INTRODUCTION

THE gap between quantum computing’s marketing and its measured performance is wide, and it is usually hidden. A tutorial runs a Bell pair or a two-qubit Grover and stops; a press release implies an “advantage” at a task someone cares about; a benchmark reports a quantum method’s accuracy without ever running the classical baseline next to it. The field’s own careful voices are clear that near-term quantum hardware is noisy and limited [1], [2] and that the headline results (quantum supremacy on a sampling task engineered to be hard for classical machines [3]) do not translate into advantage on useful problems, but that message is easy to lose. This report is an attempt to make it concrete and reproducible: to take the problems everyone learns quantum computing on,

run them on the real frameworks, put the classical baseline beside each one, and report the honest numbers.

The laboratory behind this report, QLab, is a small Problem $\times$ Solver engine. A *problem* is a formulation (search, factoring, ground-state energy, a Bell test) with its instances and observable; a *solver* is a thin adapter over a real framework: Qiskit with the Aer simulator, PennyLane, Cirq, and Stim for the quantum methods, and NumPy with scikit-learn for the classical baselines. Every run records its cost (qubits, gates, shots, wall-time) and its result, and the traces are committed so the numbers can be checked rather than believed. Over twenty problems and 119 committed traces the engine answers one question for each: run on the best real framework available, does the quantum method beat the classical baseline, and at what cost? The answer is the subject of this report, and it is the same answer the literature gives when it is honest, made measurable.

II. THE BENCHMARK: TWENTY PROBLEMS, FIVE FRAMEWORKS

The twenty problems span the standard curriculum: single-qubit and interference fundamentals; entanglement (Bell/CHSH, teleportation, superdense coding, state preparation); the oracle algorithms (Deutsch–Jozsa, Bernstein–Vazirani, Simon); the flagship algorithms (Grover’s search [4], Shor’s factoring [5], the quantum Fourier transform, quantum phase estimation); the variational methods (QAOA for MaxCut [7], VQE [8], a small quantum-machine-learning classifier); and noise and quantum error correction (a repetition code and a surface-code round, simulated with the dedicated stabilizer simulator Stim [12]). Each problem is solved by every applicable framework and by a classical baseline, under a fixed seed and shot budget, and the resulting trace records the outcome and the full cost. The costs are what make the comparison honest: a quantum method that gets the right answer is not “winning” if it needs a hundred times the wall-time of a classical routine that also gets the right answer, and the trace shows both side by side. All traces run on classical simulators of the quantum circuits, the regime a public laboratory can reach; this is stated plainly and is the correct regime for the lab-scale question the report asks.

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Technical report, 2026. Interactive laboratory and reproducible traces (MIT) at https://github.com/fsantibanezleal/CAOS_QLAB; live at <https://qlab.fasl-work.com>.

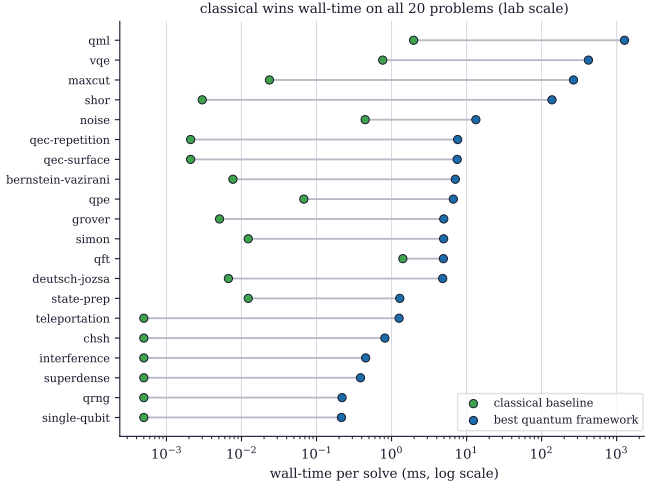


Fig. 1. Wall-time per solve, classical baseline versus the best quantum framework, on each of the twenty problems (log scale; problems sorted by quantum wall-time). The classical baseline (green) is faster on *every* problem, usually by two to four orders of magnitude, from microseconds against milliseconds on the oracle and entanglement cases to a fraction of a millisecond against hundreds of milliseconds on the variational cases (QAOA, VQE, quantum machine learning). This includes the flagship algorithms: Shor factoring 15 and Grover search are both far cheaper to run classically at this scale.

III. THE WALL-CLOCK RESULT: CLASSICAL WINS ALL TWENTY

The headline result is Fig. 1 and Table I: on all twenty problems, the classical baseline is faster in wall-clock time than the best real quantum framework, at the instance sizes a simulator reaches. The margins are large. On the entanglement and oracle problems the classical routine finishes in microseconds while the quantum framework takes single-digit milliseconds (a 100–1000 $\times$ gap dominated by circuit construction and simulation overhead). On the variational problems the gap is starkest: classical MaxCut by brute force takes 0.02 ms against QAOA’s 266 ms, and the classical VQE and quantum-machine-learning baselines finish in about a millisecond against 420 and 1274 ms for their quantum counterparts. Even the two algorithms whose entire fame is a speedup lose here: Shor’s order-finding factors 15 in 137 ms while trial division does it in 3 microseconds, and Grover’s search costs more wall-time than a classical scan of eight items. This is not a bug and not a slow implementation; it is the honest, expected consequence of asking small-instance questions, where a quantum algorithm’s asymptotic advantage is nowhere near paying for the constant-factor overhead of building and simulating a circuit.

IV. WHERE QUANTUM GENUINELY WINS: PHYSICS, NOT WALL-TIME

That classical wins the wall-clock does not mean quantum does nothing; it means the win is in physics, not practical computation, and the benchmark shows exactly where (Fig. 2). The clearest case is the CHSH/Bell test [9]. With entangled qubits and the optimal measurement settings the correlator sum reaches $S = 2.83$, the Tsirelson bound [10], and *violates* the

TABLE I
THE TWENTY PROBLEMS, REPRESENTATIVE INSTANCE: CLASSICAL VS BEST-QUANTUM WALL-TIME (MS) AND THE QUANTUM FRAMEWORK(S). CLASSICAL IS FASTER ON ALL TWENTY. FULL COSTS (QUBITS, GATES, SHOTS) ARE IN THE COMMITTED TRACES.

Problem	category	classical (ms)	quantum (ms)
single-qubit	fundamentals	< 0.001	0.22
qrng	fundamentals	< 0.001	0.22
interference	fundamentals	< 0.001	0.45
superdense	entanglement	< 0.001	0.39
teleportation	entanglement	< 0.001	1.26
chsh	entanglement	< 0.001	0.82
state-prep	entanglement	0.012	1.29
deutsch-jozsa	oracle	0.007	4.81
bernstein-vazirani	oracle	0.008	7.08
simon	oracle	0.012	4.94
grover	flagship	0.005	4.96
qft	flagship	1.42	4.91
qpe	flagship	0.068	6.66
shor	flagship	0.003	137.4
maxcut (QAOA)	variational	0.024	266.0
vqe	variational	0.76	420.6
qml	variational	1.97	1273.6
qec-repetition	noise/QEC	0.002	7.6
qec-surface	noise/QEC	0.002	7.5
noise	noise/QEC	0.45	13.3

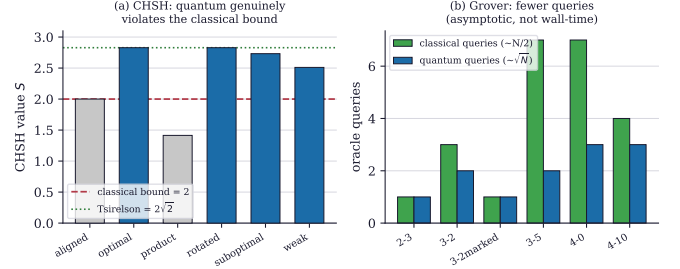


Fig. 2. The genuine quantum phenomena the benchmark demonstrates, real but not yet a wall-time advantage. (a) The CHSH value S across measurement settings: the optimal and rotated settings reach the Tsirelson bound $2\sqrt{2} \approx 2.83$ and *violate* the classical local-hidden-variable bound of 2, but only with entanglement, a product (separable) state gives $S = 1.41$ and the aligned setting sits exactly at 2. (b) Grover’s oracle-query count versus a classical scan: quantum uses provably fewer queries (two versus seven on the three-qubit, one-marked instance), the quadratic $\sqrt{N}$ advantage, which is real in query complexity even though it does not beat the classical wall-time at this N .

classical bound of 2 that any local-hidden-variable theory must obey, exactly the violation whose experimental realisation [11] earned the 2022 Nobel Prize. This is a genuine quantum-beats-classical result, but it is a statement about the structure of nature (local realism is false), not a faster computation, and it requires entanglement: the benchmark’s product-state variant gives $S = 1.41$ and does not violate anything. The second case is query complexity: Grover’s algorithm [4] finds a marked item in about $\sqrt{N}$ oracle queries against a classical scan’s $N/2$, and the benchmark measures the advantage directly (two quantum queries against seven classical on one instance). This too is real, a proven separation in the number of oracle calls, and it too fails to beat wall-time at reachable N because each quantum query costs far more to simulate than a classical array lookup. Shor’s algorithm [5] completes the picture: it factors 15 correctly through order-finding, a correct execution of a genuinely quantum algorithm, while remaining, at the scale that matters, no threat, factoring RSA-2048 would need on

the order of a million noisy physical qubits with full fault tolerance [6], which is why the honest verdict on the trace is that Shor is not a near-term cryptographic danger.

V. DISCUSSION

The benchmark separates two questions the marketing conflates. *Does quantum mechanics do things classical physics cannot?* Yes, and the CHSH violation is the clean demonstration. *Does a quantum computer beat a classical one at a task you would pay for, today, at the scale we can reach?* No, on all twenty problems, and the wall-times say so by two to four orders of magnitude. The two answers are not in tension; they are the honest state of the field, and the value of putting them side by side is that a learner sees both the genuine physics and the practical reality without either being hidden. The report also makes a methodological point: a quantum-vs-classical claim is only meaningful with the classical baseline actually run and its cost reported. A great deal of published quantum “advantage” evaporates under that discipline, either because the classical baseline was never run, or because the advantage is asymptotic and the reported instance is far too small for it to appear. The benchmark builds the discipline into the architecture: every quantum solver has a classical sibling, and both costs are always on screen.

VI. SCOPE AND LIMITATIONS

The scope is explicit. Every trace runs on a *classical simulator* of the quantum circuit, not on quantum hardware; this is the regime a public laboratory can reach and is the correct regime for the lab-scale question, but it means the wall-times measure simulation cost, not a physical quantum processor, and a fault-tolerant machine at large scale would change the asymptotic story for Shor and for quantum chemistry (not demonstrated here, and not claimed). The instances are small (a handful of qubits), by necessity, so the benchmark measures the small- N regime where asymptotic advantages are absent by construction; it does not and cannot refute the existence of a large- N advantage, only show that it is not present at lab scale. The noise and error-correction cases are idealised-to-realistic simulator models, not device calibration data. And the frameworks are pinned versions; a newer release could shift the constants. None of this is hidden: the laboratory states the same scope on every panel, and the point of the report is the honest small-scale record, not a claim about the ultimate limits of quantum computing.

VII. CONCLUSION

Run honestly, with the classical baseline beside every quantum method and both costs on the table, twenty canonical quantum problems on five real frameworks give one answer: at lab scale the classical baseline wins the wall-clock on all twenty, including the flagship speedup algorithms, because their advantages are asymptotic and the instances are small. And yet quantum genuinely wins at physics: the CHSH bound is violated, Grover’s query advantage is real, Shor factors correctly. The honest synthesis, that quantum beats classical at nonlocality and query complexity but not yet at any practical

computation you would pay for, is not a disappointment; it is the measured truth, and a benchmark that shows it reproducibly, with the numbers over real frameworks, is worth more to a learner than another toy circuit that implies an advantage it cannot demonstrate.

APPENDIX A REPRODUCTION

The benchmark is the 119 traces committed under `data/artifacts/`, one folder per problem, each trace recording the problem, the solvers, the per-solver costs, and the honest quantum-vs-classical verdict. The aggregation script (`manuscripts/quantum-vs-classical/figures/aggregate`) walks them into the summary artifact the figures read, extracting the per-problem wall-times, the CHSH value across settings, and the Grover query counts. Re-running the traces (each is seeded) reproduces the numbers; the figures and Table I regenerate from the committed summary.

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